

Problem 10.40

Show that

$$\int_0^n \delta_t dt = \ln a(n).$$

Solution

By definition, the force of interest is

$$\delta_t = \frac{a'(t)}{a(t)}.$$

Therefore,

$$\begin{aligned} \int_0^n \delta_t dt &= \int_0^n \frac{a'(t)}{a(t)} dt \\ &= \ln a(t) \Big|_0^n \\ &= \ln a(n) - \ln a(0). \end{aligned}$$

Since the accumulation function satisfies $a(0) = 1$,

$$\ln a(0) = \ln(1) = 0.$$

Hence,

$$\boxed{\int_0^n \delta_t dt = \ln a(n)}.$$